

Team Project Reflection

What?

The initial team meetings were instrumental in defining the direction and scope of the project. During these sessions, we developed a preliminary project plan, established key objectives, and allocated responsibilities according to each member's strengths and expertise. Two priority tasks were identified at the outset: formulating the business problem and conducting exploratory data analysis (EDA). Ross focused on defining the target variable in alignment with the project's primary objective of identifying the key drivers of Airbnb demand, while Sonya led the EDA process to better understand the dataset and identify potential modelling opportunities.

Following this foundational stage, the project moved into model development. As a team, we explored three complementary deep learning approaches designed to capture different forms of information within the dataset. These included a **tabular neural network** for structured numerical and categorical data, a **sentiment analysis model** for text-based features, and a **convolutional neural network (CNN)** for image patches generated from geographic coordinates. After collectively agreeing on the modelling framework and overall methodology, each team member independently developed and evaluated their assigned model.

The next phase involved integrating these standalone models into a unified analytical framework and comparing different model configurations to identify the most effective solution. Performance evaluation played a crucial role throughout this stage. Metrics such as **accuracy**, **ROC-AUC**, and **learning curves** enabled us to assess predictive performance, identify potential weaknesses, and compare alternative approaches objectively. The final stage focused on interpreting the results and translating technical findings into meaningful business recommendations for Airbnb. This was a highly collaborative process that required combining technical analysis with business understanding to generate practical and actionable insights.

So What?

My primary responsibility within the project was the development of the **CNN model**, which proved to be one of the most challenging and rewarding aspects of the project. One of the initial difficulties involved selecting an appropriate modelling strategy. Unlike previous projects, which had largely focused on structured tabular data, this task involved working with **unstructured image data** derived from geographic locations. This required me to rethink how information could be represented and processed, as image-based models rely on fundamentally different learning mechanisms than traditional machine learning algorithms.

A second challenge was **model interpretability**. While interpretability remains possible in CNN models, it is considerably less straightforward than in traditional statistical methods such as linear

regression or decision trees. To better understand model behaviour, I explored techniques such as **feature maps**, which illustrate where the network detects patterns within images, and **heatmaps**, which highlight the regions of an image that contribute most strongly to a prediction. These techniques provided useful insights into the model's decision-making process, but they were inherently less transparent than the direct coefficient-based interpretations available in simpler models.

This experience reinforced an important lesson: as predictive models become more sophisticated, interpretability often becomes more challenging. Consequently, I gained a greater appreciation for the importance of **post-hoc interpretability techniques**, which can help explain model outputs and increase stakeholder trust in complex machine learning systems.

From a teamwork perspective, the project was an extremely positive learning experience. One of the greatest strengths of the group was the culture of mutual respect and collaboration. Every member's contribution was valued equally, creating an environment that encouraged creativity, experimentation, and open discussion. This inclusive atmosphere increased confidence within the team and enabled us to explore innovative ideas without hesitation.

Another factor contributing to the project's success was our pragmatic approach. Given the time constraints, we focused on delivering meaningful results efficiently rather than becoming overly focused on minor details that would have limited overall progress. This mindset ensured that effort was directed towards activities that added the greatest value to the final deliverable.

Effective task allocation also played a critical role in the project's success. Team members with technical and computing backgrounds took responsibility for advanced modelling tasks such as CNN development and sentiment analysis, while those with stronger business and humanities backgrounds focused on problem definition, business interpretation, and the development of actionable recommendations. This division of labour enabled the team to leverage individual strengths effectively and resulted in a final report that achieved a strong balance between **technical rigor and business relevance**.

Now What?

One of the most valuable outcomes of this project was gaining practical experience with two of the most widely used deep learning frameworks: **TensorFlow/Keras** and **PyTorch**. Unlike traditional machine learning libraries, these frameworks support the entire machine learning lifecycle, from model development and training to evaluation and deployment.

When selecting a framework for the CNN model, I evaluated several factors, including programming style, usability, and ecosystem support. I found **TensorFlow with Keras** particularly appealing because its high-level, object-oriented API allows code to be written in a more readable and maintainable manner. **PyTorch** also offers significant advantages, particularly its seamless integration with native Python programming structures and its flexibility during experimentation. However, it

generally requires more manual implementation compared to Keras, which can increase development time for certain applications.

From an ecosystem perspective, both frameworks are highly mature and well-supported. PyTorch benefits from a rapidly growing research community, while TensorFlow offers a broader suite of production-ready tools and resources. Ultimately, I chose TensorFlow because of its strong **observability and monitoring capabilities**, particularly **TensorBoard**, which allowed me to visualize training progress, diagnose issues, and better understand model behaviour throughout development.

Beyond the technical knowledge gained, this project strengthened my ability to be **adaptable and resourceful**. Throughout the project, we encountered constraints related to tools, data availability, and design decisions. Learning to work effectively within these limitations and maximize the value of available resources was an important professional skill. I realized that successful data science projects often depend as much on adaptability and practical decision-making as they do on technical expertise.

Looking ahead, I would like to continue improving the CNN approach developed in this project. In particular, I see potential in enhancing geographic image patches with **engineered spatial features** that explicitly capture location-specific characteristics such as accessibility, neighbourhood attributes, points of interest, and local activity levels. By incorporating these additional signals, the model could gain a richer understanding of the spatial environment and potentially improve predictive performance.

Overall, this project significantly enhanced my technical capabilities in deep learning, strengthened my understanding of model interpretability, and reinforced the value of effective collaboration in multidisciplinary teams. It also highlighted the importance of adaptability, thoughtful model design, and balancing technical sophistication with practical business value—skills that I intend to apply in future machine learning and data science projects.